

Two Waves Can Add Up To Nothing

SYNOPSIS

In Note 3 you learned that a single wave is three numbers and a phase: amplitude A , wavelength λ , frequency f , tied together by $v = \lambda f$ — and **phase**, the idea you only need once you have *two* waves to compare. This note is where that fourth idea earns its keep. Everything so far has been about one wave travelling on its own. What makes waves genuinely different from the rocks and baseballs of the rest of physics is what happens when two of them arrive at the same place at the same time. Particles collide and bounce. Waves do something no particle can do: they **interfere**. They add where they arrive in step, and they *subtract* where they arrive out of step — so completely that two loud sounds can produce silence. Interference, along with diffraction and refraction, is on the short list of behaviours that forced physicists to build a whole separate description for waving things. We start with interference, and we start with your own ears.

BEFORE YOU READ ON — COMMIT TO AN ANSWER

Two people hold the ends of a long slinky stretched across the floor. At the same instant, each of them flicks their end *upward* and sends a pulse travelling toward the other. The two pulses race toward each other and meet in the middle. **At the moment they meet, and just afterwards, what happens?**

- ☐ They bounce off each other and each travels back the way it came.
- ☐ They briefly pile into one big pulse, then separate and keep going, each one unchanged.
- ☐ They merge permanently into a single pulse that continues in one direction.
- ☐ They cancel and both disappear.

Now the harder one. Run it again, but this time one person flicks *up* and the other flicks *down*. For one instant as they overlap, the slinky lies completely flat — no pulse anywhere. Write one sentence saying where the energy is at that instant.

Particles bounce. Waves pass through.

You already know what two baseballs do when they meet: they collide, and after the collision neither one is doing what it was doing before. Two waves meeting in the same medium do not collide at all. Each one simply keeps going as though the other were not there. The disturbance you see *while* they overlap is nothing more than the two disturbances added together, point by point, and once they have passed through each other both emerge with their original shape and speed intact. This is the **principle of superposition**, and it is the whole of this note in one sentence:

$$\text{total displacement} = \text{displacement from wave 1} + \text{displacement from wave 2}$$

Holds wherever two waves share a medium and the disturbances are not enormous. **Note the sign:** a displacement below equilibrium is negative, so “adding” can make the total smaller — or zero.

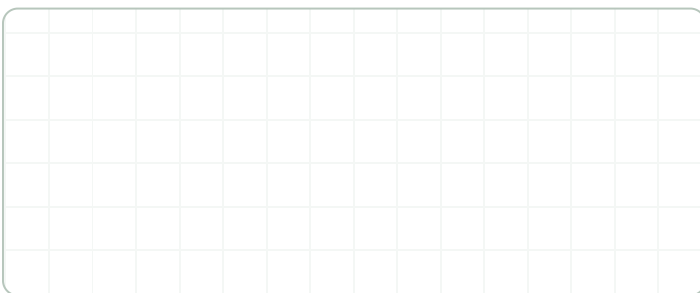
That parenthetical is the entire surprise. Because displacements carry a sign, adding two waves can produce *less* than either one alone. Nothing in your experience of colliding objects prepares you for two things arriving somewhere and the result being nothing. It happens only because a wave is a disturbance from equilibrium rather than a piece of stuff, and a disturbance can point either way.

And notice what the flat-slinky moment tells you about energy. The slinky is flat for an instant, but it is not *still*: every coil in that region is moving at maximum speed, passing through equilibrium on its way to the other side. The energy never left. It moved out of the shape of the medium and into the motion of the medium, and an instant later it moves back. Destructive interference relocates energy; it never destroys it.

Two ways for waves to add

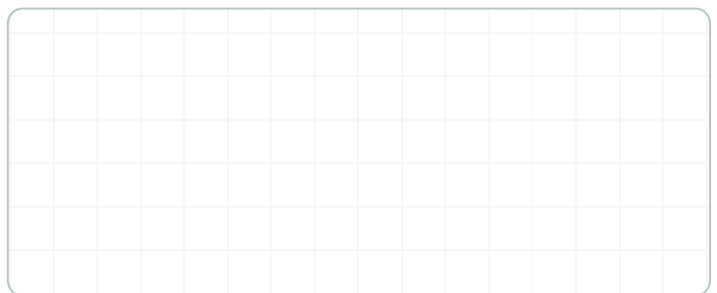
When two identical waves overlap, there are two extreme cases worth naming, and everything else lies between them. Draw each one below: sketch the two waves lightly, then draw the sum darker on top.

Constructive interference



Describe in words:

Destructive interference



Describe in words:

Constructive interference is crest landing on crest and trough on trough: the two waves are **in phase**, the displacements have the same sign everywhere, and the resultant amplitude is the sum of the two. Destructive interference is crest landing on trough: the two waves are **180° out of phase**, the signs are opposite everywhere, and the resultant amplitude is the difference. For two *identical* waves that difference is zero — perfect cancellation.

CAREFUL

Interference changes the **amplitude** of the result. It does not change the frequency, and it does not change the wave speed. Two 800 Hz waves interfering give you a louder or quieter 800 Hz — never a different pitch.

Phase is set by distance

So far this is a statement about two waves that are already in step or already opposed. The useful question is the one you can actually control: *given two sources, where in the room do you get each case?*

Suppose two speakers play the identical tone, and both push air outward at the same instant — they are **in phase at the source**. Stand somewhere and ask what each wave has done on its way to you. A wave advances one full cycle for every wavelength it travels. So if the two paths to your ear are the same length, both waves have turned through the same number of cycles and arrive in step. If one path is longer by exactly one wavelength, that wave has turned through one extra *whole* cycle — which puts it back in step again. Two extra wavelengths, still in step. But if one path is longer by *half* a wavelength, that wave arrives exactly opposed, and you get cancellation.

Write the difference in the two path lengths as $\Delta = |r_A - r_B|$. Then:

$$\Delta = n\lambda \rightarrow \text{constructive (LOUD)}$$

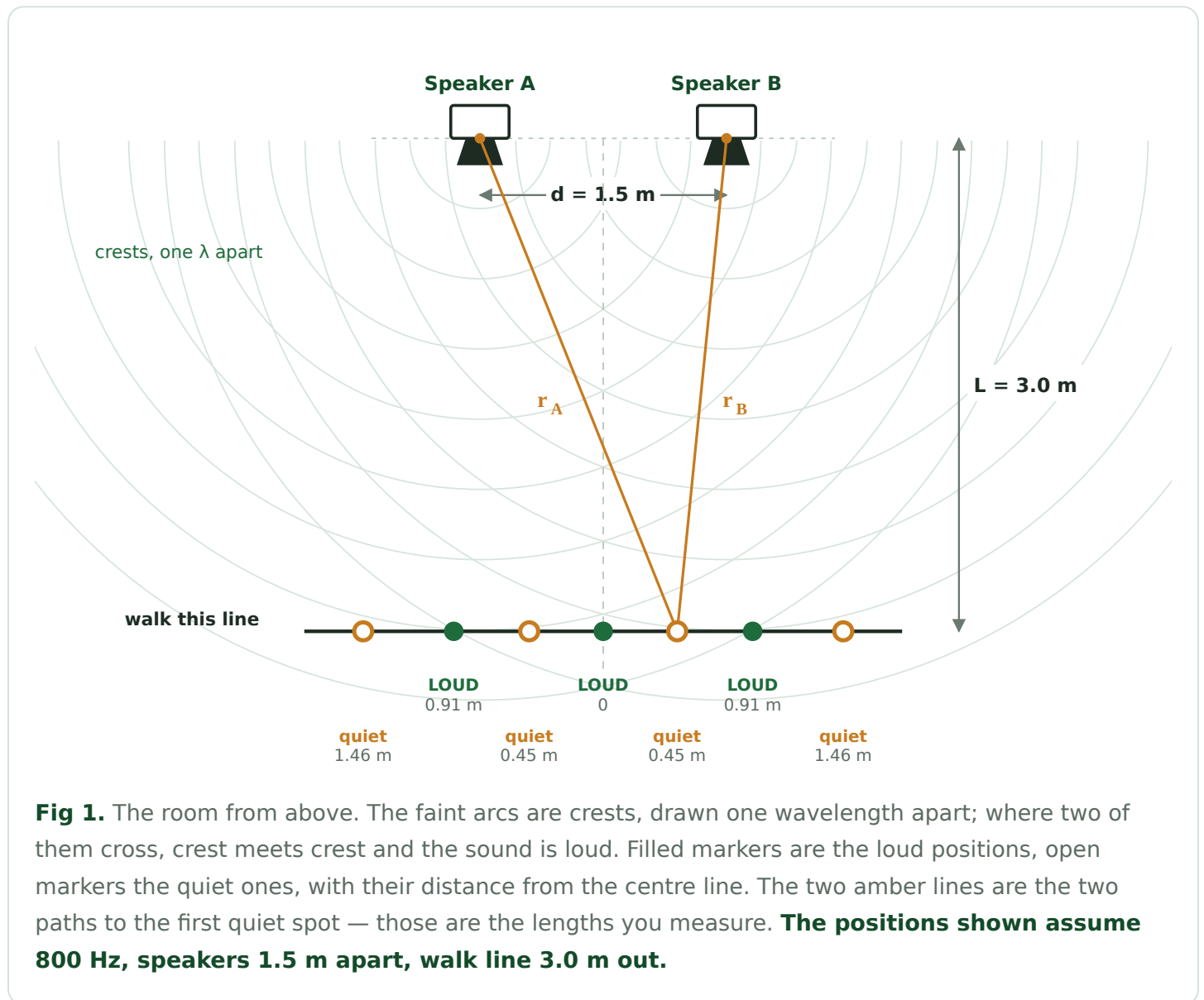
$$\Delta = (n + \frac{1}{2})\lambda \rightarrow \text{destructive (quiet)}$$

$n = 0, 1, 2, \dots$ · **Valid only when the two sources emit in phase with each other, at the same frequency, into the same medium.** Reverse the wires on one speaker and the two results swap places. Different media — water on one side, air on the other — and λ is not the same on the two paths, so this form fails.

Read the two lines together and notice how economical this is. You do not need to know the amplitude, the loudness, or how far away the speakers are. One length — the *difference* of two distances — measured against one other length, the wavelength, decides whether a spot in the room is loud or quiet. That is the whole rule, and you are about to check it with a tape measure.

The two-speaker walk

Two speakers on the front bench play the same steady 800 Hz tone. The sound fills the room, but it does not fill it *evenly*. Walk slowly along a line a few metres out, with one ear plugged, and you will hear the tone rise and fall — loud, quiet, loud, quiet — even though nothing about the speakers is changing. Those quiet spots are not the sound failing to reach you. They are two sounds arriving and cancelling.



Two details about the procedure are worth understanding rather than just obeying. **Plug one ear.** Your two ears are about 18 cm apart, and at 800 Hz the wavelength is only 43 cm, so your two ears sample two genuinely different points in the pattern and your brain averages them into mush. One ear gives you one point. **And walk slowly.** The quiet spots are dips of perhaps ten or twenty decibels, not silences — real rooms have reflections that partly fill them in. You are listening for a clear minimum, not for nothing at all.

Position on the line	Loud or quiet?	r_A (m)	r_B (m)	$\Delta = r_A - r_B $ (m)	$\Delta \div \lambda$
centre (0 m)					
first quiet spot					
next loud spot					
second quiet spot					

Performance Expectations

State the principle of superposition and use it to explain why two waves pass through each other unchanged while two particles do not. Distinguish constructive from destructive interference by the phase relationship between the two waves, and describe the resultant amplitude in each case. Explain where the energy is at an instant of complete destructive interference. Predict whether a given point is loud or quiet from the difference in path length to two in-phase sources, using $\Delta = n\lambda$ and $\Delta = (n + \frac{1}{2})\lambda$. Measure two path lengths in a real room, compute their difference, and compare it with the wavelength obtained from $v = \lambda f$. State the conditions under which those two relations hold.

Instructions

Answer questions 1–6 from the note itself. Questions 7–12 use the numbers your class measured during the speaker walk — fill in the table above first. Take the speed of sound as $v = 343$ m/s.

- 1

In your own words, what does a **particle** do when it meets another particle, and what does a **wave** do when it meets another wave? Name the one property of a wave that makes the difference possible.
- 2

Two identical pulses, both upward, meet on a slinky. Sketch the medium at the instant they fully overlap, and state the amplitude of what you drew in terms of the amplitude A of one pulse.

Resultant amplitude = _____

3

Now one pulse is upward and one is downward, otherwise identical. Sketch the medium at the instant they fully overlap. Then answer the question from the top of this note: **where is the energy** at that instant? Your answer must not contain the word “destroyed.”

4

A tone at 800 Hz travels through air at 343 m/s. Find its wavelength. **State the equation first**, then substitute.

$\lambda =$ _____ (equation)

$\lambda =$ _____ m

5

Using your answer to question 4, what path difference Δ produces the **first** quiet spot away from the centre line? And the **first** loud spot away from the centre line?

first quiet: $\Delta =$ _____ m

first loud (off centre): $\Delta =$ _____ m

6

Explain *why* the point exactly halfway between the two speakers is always loud, without doing any arithmetic. What is Δ there, and what does that say about the two waves?

7

From your table: at the first quiet spot, what was Δ ? Divide it by the wavelength you found in question 4. How close is the result to $\frac{1}{2}$?

$\Delta \div \lambda =$ _____

8

From your table: at the off-centre loud spot, what was $\Delta \div \lambda$? How close is the result to a whole number?

$\Delta \div \lambda =$ _____

9

Turn the measurement around. Suppose you did *not* know the frequency. Using only your measured Δ at the first quiet spot and the fact that it should equal $\lambda/2$, compute λ , then compute f from $v = \lambda f$. How close is it to 800 Hz?

λ from measurement = _____ m

f = _____ Hz

- 10** Name two things about the real room that would make your measured Δ differ from the ideal value. For each one, say whether it would make the quiet spots harder to *hear* or harder to *locate*.
- 11** Predict: if the tone were changed from 800 Hz to 1600 Hz and nothing else moved, what happens to the spacing between the quiet spots along the walk line? Justify your answer from $\Delta = (n + \frac{1}{2})\lambda$, not from memory.
- 12** Predict: if the two speakers were moved *closer together* and the tone left alone, what happens to the spacing between the quiet spots? (You will test both of these predictions in the next note.)

WHERE THIS IS GOING

You now have a rule — $\Delta = n\lambda$ for loud, $\Delta = (n + \frac{1}{2})\lambda$ for quiet — and one room's worth of evidence for it, taken with your own ears and a tape measure. One room is not a lot of evidence, and a classroom is a bad laboratory: you cannot turn off the walls, you cannot change the speed of sound, and you certainly cannot see the pattern. In Note 4b you take the same rule to a simulation where you *can* see it, in two different media at once — ripples on water and sound in air — and where the frequency and the source separation are knobs you turn. The predictions you just made in questions 11 and 12 are the first things you will check.

Files for this note

Files for this note. The read PDF and the printable handout live at
byungmooncho.github.io/PHY105/handouts/M1_N4a_Wave_Interference.pdf
byungmooncho.github.io/PHY105/handouts/M1_N4a_Wave_Interference_Handout.pdf